Mixing angle between $^3P_1$ and $^1P_1$ states in heavy axial vector mesons within the QCD sum rules framework

T. M. Aliev,[1,*] S. Bilmis,[1,2,†] and M. Savci[1,‡]

[1]*Department of Physics, Middle East Technical University, Ankara, 06800, Turkey*
[2]*TUBITAK ULAKBIM, Ankara, 06510, Turkey*

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In this study, we calculate the mixing angles between the axial-vector mesons $D_{1(s1)} - D'_{1(s1)}$ and $B_{1(s1)} - B'_{1(s1)}$ using the QCD sum rules approach. Our results are $\theta_1 = 28.2 \pm 0.6°$, $\theta_2 = 26.6 \pm 0.6°$, $\theta_3 = 38.6 \pm 0.1°$, and $\theta_4 = 38.5 \pm 0.1°$. These values are in good agreement with the predictions of heavy quark effective theory, particularly for the mixing angle $\theta = 35.3°$, and are compatible with several existing results in the literature. The predicted mixing angles can be tested through the analysis of semileptonic decays such as $B_c \to B_1 \ell\nu$, $B_c \to B^0_{s1}\ell\nu$, $B_s \to D_{s1}\ell\nu$, and $B_s \to D'_{s1}\ell\nu$, which can be investigated at experimental facilities such as LHCb and Belle II.

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I. INTRODUCTION

The quark model has been very successful in the classification of hadrons. According to the quark model, axial-vector mesons with quantum numbers $J^P = 1^+$ are grouped into a nonet. In spectroscopic notation $n^{2S+1}L_j$, there are two types of lowest p-wave mesons: 1^3P_1 and 1^1P_1 with C-parity $C = +1$ and $C = -1$, respectively. These states are usually denoted as A_A and A_B. The physical mass eigenstates are mixtures of A_A and A_B states. The physical states A and A' are defined in terms of A_A and A_B as follows:

$$A = A_A \sin\theta + A_B \cos\theta,$$
$$A' = A_A \cos\theta - A_B \sin\theta, \tag{1}$$

where θ is the mixing angle between A_A and A_B states.

In this study, we focus our attention on charmed and bottom mesons with states $J^P = 1^+$. Specifically, we consider the mixing between $D_1 - D'_1(\theta_1)$, $D_{s1} - D'_{s1}(\theta_2)$, $B_1 - B'_1(\theta_3)$, and $B_{s1} - B'_{s1}(\theta_4)$, which are listed with their associated masses in Table I. Note that although B_1 and B_{s1} states have not been discovered yet, different theoretical approaches predicted slightly smaller mass splittings of 10–30 MeV between B_1 and B'_1 (see [1] and references therein).

The mixing angle is fundamental not only for understanding the nature of heavy axial-vector mesons but also for accurately determining their decay widths, which are critical for both theoretical studies and experimental verification.

One of the earliest comprehensive analyses of the mixing angles for $L = 1$ mesons was conducted using the relativistic quark model [3]. Since then, the mixing angle θ has been studied extensively using alternative approaches, including Bethe-Salpeter method [4], relativistic quark models [3,5–8], nonrelativistic quark models [1,9], QCD potential models [10], constituent quark model [11,12], chiral quark model [13], Coulomb-gauge QCD model [14], $^3P_0$ pair creation model [15], and strong decay model [16] (see also review paper [17]).

TABLE I. The heavy-light mesons with $J^P = 1^+$ [2]. For the undiscovered states B'_1 and B'_{s1}, the mass values are adapted from theoretical predictions [1].

State-pair	States	Mass (MeV)
$D_1 - D'_1(\theta_1)$	$D_1(2420)$	2421.4 ± 0.6
	$D_1(2430)$	2427 ± 36
$D_{s1} - D'_{s1}(\theta_2)$	$D_{s1}(2460)$	2459.5 ± 0.6
	$D_{s1}(2536)$	2535.1 ± 0.1
$B_1 - B'_1(\theta_3)$	$B_1(5710)$	5710
	$B_1(5721)$	5726.1 ± 1.3
$B_{s1} - B'_{s1}(\theta_4)$	$B_{s1}(5820)$	5820
	$B_{s1}(5830)$	5828.7 ± 0.4

*Contact author: taliev@metu.edu.tr
†Contact author: sbilmis@metu.edu.tr
‡Contact author: savci@metu.edu.tr

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In this work, we calculate the mixing angles θ using the QCD sum rules approach, following the methodology first introduced in [18]. The paper is organized as follows. In Sec. II, we derive the sum rules for the mixing angles of 1^3P_1 and 1^1P_1 states for heavy axial-vector mesons. In Sec. III, we perform a numerical analysis of the sum rules for the mixing angles, and the final section contains our conclusion.

II. THEORETICAL FRAMEWORK AND DERIVATION OF MIXING ANGLES

In general, mass eigenstates do not coincide with flavor eigenstates. Hence, the mass eigenstates can be represented as linear combinations of flavor eigenstates. This implies that the interpolating currents for A_i and A_i' mesons can be described as linear combinations of the currents for flavor eigenstates, i.e.,

$$\begin{aligned} J_{A_i\mu} &= \sin\theta_i J^{(0)}_{A_i\mu} + \cos\theta_i J^{(0)}_{B_i\mu}, \\ J_{A_i'\mu} &= \cos\theta_i J^{(0)}_{A_i\mu} - \sin\theta_i J^{(0)}_{B_i\mu}, \end{aligned} \tag{2}$$

where the superscript (0) denotes the current in the flavor eigenstates and

$$\begin{aligned} J^{(0)}_{A_\mu} &= \bar{q}\gamma_\mu\gamma_5 Q, \\ J^{(0)}_{B_\nu} &= i\bar{q}\sigma_{\nu\alpha}p^\alpha\gamma_5 Q, \end{aligned} \tag{3}$$

with q and Q representing the light and heavy quarks, respectively. Here $i = 1, 2, 3, 4$ correspond to the axial-vector D_1, D_{s1}, B_1, and B_{s1} states.

To determine the mixing angles θ_i, we follow the method presented in [18] and start by considering the correlation function:

$$\Pi^{(AA')}_{\mu\nu}(p) = \int d^4x e^{ipx}\langle 0|T\{J_{A_\mu}(x)\bar{J}'_{A'_\nu}(0)\}|0\rangle. \tag{4}$$

To find the sum rules for the mixing angle θ_i, the correlation function is evaluated in two different domains. On one side, it is calculated in terms of hadrons by saturating a full set of hadrons carrying the same quantum numbers as the interpolating currents. On the other side, the correlation function is calculated in the deep Euclidean region ($p^2 \ll 0$) using the operator product expansion (OPE). These two representations are then matched, and to suppress higher states and the continuum, a Borel transformation with respect to the variable $-p^2$ is performed. Since the currents $J_{A\mu}$ and $J'_{A'\nu}$ create only A and A' mesons from the vacuum, the hadronic part of the correlation function vanishes. Using Eqs. (2) and (3) from Eq. (4) we get

$$\cos\theta\sin\theta(\Pi^{(0)AA}_{\mu\nu} - \Pi^{0BB}_{\mu\nu}) + (\cos^2\theta - \sin^2\theta)\Pi^{(0)AB}_{\mu\nu} = 0, \tag{5}$$

where $\Pi^{(0)ij}$ are the correlation functions for the unmixed states, i.e.,

$$\Pi^{(0)ij}_{\mu\nu} = i\int d^4x\, e^{ipx}\langle 0|T\{J^{(0)i}_\mu(x)\bar{J}^{(0)j}_\nu\}|0\rangle. \tag{6}$$

For axial-vector current, the correlation function can be expressed in terms of two independent invariant structures:

$$\Pi^{(0)ij}_{\mu\nu}(p^2) = \Pi^{ij}_1(p^2)\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right) + \Pi^{ij}_2(p^2)\frac{p_\mu p_\nu}{p^2}. \tag{7}$$

The structure $g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}$ is associated with spin-1 particles. Thus, we consider only this structure. Extracting the coefficient of this structure from Eq. (5), the mixing angles are finally determined as

$$\tan 2\theta = -\frac{2\Pi^{AB}_1}{\Pi^{AA}_1 - \Pi^{BB}_1}. \tag{8}$$

It is worth noting that different conventions for the sign of mixing angles are used in the literature. This variation arises from the choice of quark ordering in representing the heavy-light quark system, which can be written either as $\bar{q}Q$ or $\bar{Q}q$. This difference in representation leads to opposite signs for the mixing angles. In our analysis, we adopt the convention presented in [19], where the system is written as $Q\bar{q}$.

With the conventions defined, we proceed to calculate the theoretical part of the correlation function in the deep Euclidean region ($p^2 \ll 0$) using the OPE. The explicit expressions for the interpolating currents are employed, and Wick's theorem is applied. After this operation, the correlation function is obtained in terms of quark propagators as follows:

$$\begin{aligned} \Pi^{(0)AA}_{\mu\nu}(p^2) &= -i\int d^4x\, e^{ipx}\mathrm{Tr}[S^{ab}_q(-x)\gamma_\mu\gamma_5 S^{ba}_Q(x)\gamma_\nu\gamma_5], \\ \Pi^{(0)AB}_{\mu\nu}(p^2) &= -i\int d^4x\, e^{ipx}\mathrm{Tr}[S^{ab}_q(-x)\sigma_{\mu\alpha}p^\alpha\gamma_5 S^{ba}_Q(x)\gamma_\nu\gamma_5], \\ \Pi^{(0)BB}_{\mu\nu}(p^2) &= -i\int d^4x\, e^{ipx}\mathrm{Tr}[S^{ab}_q(-x)\sigma_{\mu\alpha}p^\alpha\gamma_5 \\ &\quad \times S^{ba}_Q(x)\sigma_{\nu\beta}p^\beta\gamma_5]. \end{aligned} \tag{9}$$

The quark propagators for the light and heavy quarks in the x representation are given by Eqs. (9) and (10), respectively, where $G^{\mu\nu}$ is the gluon field strength tensor, and K_i are the modified Bessel functions of the second kind (see [2,20]),

$$iS_q^{ab}(x) = \frac{i\delta^{ab}}{2\pi^2x^4}\not{x} - \frac{\delta^{ab}}{12}\langle\bar{q}q\rangle - \frac{i\delta^{ab}g_s^2x^2\not{x}}{2^5 3^5}\langle\bar{q}q\rangle^2 + \frac{i}{32\pi^2}g_sG_{\mu\nu}^{ab}\frac{\sigma^{\mu\nu}\not{x}+\not{x}\sigma^{\mu\nu}}{x^2} + \frac{\delta^{ab}x^2}{192}\langle g_s\bar{q}\sigma Gq\rangle$$
$$- \frac{\delta^{ab}x^4}{2^{10}3^3}\langle\bar{q}q\rangle\langle g_s^2G^2\rangle + \cdots - \frac{m_q\delta^{ab}}{4\pi^2x^2} + \frac{im_q\delta^{ab}\not{x}}{48}\langle\bar{q}q\rangle - \frac{m_q\delta^{ab}g_s^2x^4}{2^7 3^5}\langle\bar{q}q\rangle^2$$
$$+ \frac{m_q}{32\pi^2}g_sG_{\mu\nu}^{ab}\sigma^{\mu\nu}\ln(-x^2) - \frac{im_q\delta^{ab}x^2\not{x}}{2^7 3^2}\langle g_s\bar{q}\sigma Gq\rangle - \frac{m_q\delta^{ab}}{2^9 3\pi^2}x^2\ln(-x^2)\langle g_s^2G^2\rangle + \cdots, \tag{10}$$

$$S_Q^{ab}(x) = \frac{m_Q^2\delta^{ab}}{(2\pi)^2}\left[i\not{x}\frac{K_2\left(m_Q\sqrt{-x^2}\right)}{(\sqrt{-x^2})^2} + \frac{K_1\left(m_Q\sqrt{-x^2}\right)}{\sqrt{-x^2}}\right] - \frac{m_Qg_sG_{\mu\nu}^{ab}}{8(2\pi)^2}\left[i(\sigma^{\mu\nu}\not{x}+\not{x}\sigma^{\mu\nu})\frac{K_1\left(m_Q\sqrt{-x^2}\right)}{\sqrt{-x^2}} + 2\sigma^{\mu\nu}K_0\left(m_Q\sqrt{-x^2}\right)\right]$$
$$- \frac{\langle g_s^2G^2\rangle\delta^{ab}}{2^6 3^2(2\pi)^2}\left[(im_Q\not{x}-6)\frac{K_1\left(m_Q\sqrt{-x^2}\right)}{\sqrt{-x^2}} + m_Qx^4\frac{K_2\left(m_Q\sqrt{-x^2}\right)}{(\sqrt{-x^2})^2}\right]$$
$$+ \frac{\langle g_s^3G^3\rangle\delta^{ab}}{2^8 3^2(2\pi)^2}\left[-\frac{i\not{x}x^2}{m_Q}\frac{K_1\left(m_Q\sqrt{-x^2}\right)}{\sqrt{-x^2}} + i\not{x}x^4\frac{K_2\left(m_Q\sqrt{-x^2}\right)}{(\sqrt{-x^2})^2}\right.$$
$$\left. + \frac{10}{m_Q}x^4\frac{K_2\left(m_Q\sqrt{-x^2}\right)}{(\sqrt{-x^2})^2} + x^4\frac{K_1\left(m_Q\sqrt{-x^2}\right)}{\sqrt{-x^2}}\right]. \tag{11}$$

The invariant function for the Lorentz structure $g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}$ can be expressed with its imaginary part (spectral density) via the dispersion relation:

$$\Pi_1^{ij}(p^2) = \int_{m_Q^2}^{\infty} ds\,\frac{\rho_1^{ij}(s)}{s-p^2}. \tag{12}$$

Using the explicit expressions for the light and heavy quark propagators, the spectral density can be calculated straightforwardly. The expressions for the spectral densities are presented in the Appendix. After performing Borel transformation over the variable $(-p^2)$ and imposing quark-hadron duality, we get

$$\Pi_1^{ij}(M^2) = \int_{m_Q^2}^{s_0} ds\rho_1^{ij}(s)e^{-s/M^2}, \tag{13}$$

where s_0 is the continuum threshold in the corresponding channel. Finally, substituting these results into Eq. (8), we can determine the mixing angle.

III. NUMERICAL ANALYSIS

After having established the theoretical framework, we conduct a numerical analysis of the sum rules to determine the mixing angle in this section. We begin by listing the input parameters used in the sum rules in Table II. The heavy quark masses are given in the $\overline{\text{MS}}$ scheme, while the values of strange quark mass and the quark condensate are presented at $\mu = 1$ GeV scale.

In addition to these input parameters, there are two auxiliary parameters in the sum rules: the Borel mass parameter M^2 and the continuum threshold s_0. Table III summarizes the working regions of these parameters, which are chosen to ensure the stability and convergence of the sum rules.

Since M^2 and s_0 are auxiliary parameters, the mixing angle should be independent of them. The working region of M^2, where the mixing angle is very weakly dependent on these parameters, is determined by imposing specific conditions.

The upper limit of M^2 is determined by demanding that higher state and continuum contributions remain below

TABLE II. The numerical values of the input parameters.

$\bar{m}_s(1\text{ GeV})$	0.126 GeV [2]
$\bar{m}_b(\bar{m}_b)$	$4.18^{+0.03}_{-0.02}$ GeV [2]
$\bar{m}_c(\bar{m}_c)$	(1.27 ± 0.02) GeV [2]
$\langle\bar{q}q\rangle(1\text{ GeV})$	$(-1.65 \pm 0.15) \times 10^{-2}$ GeV^3 [21]
$\langle\bar{s}s\rangle$	$(0.8 \pm 0.2)\langle\bar{q}q\rangle$ GeV^3 [21]
m_0^2	(0.8 ± 0.2) GeV^2 [21]
$\langle g^2G^2\rangle$	$4\pi^2(0.012 \pm 0.006)$ GeV^4 [22]
$\langle g^3G^3\rangle$	(0.57 ± 0.29) GeV^6 [23]

TABLE III. The working regions for the Borel mass parameter M^2 and the continuum threshold s_0.

	M^2 (GeV^2)	s_0 (GeV^2)
D_1	$3 \le M^2 \le 6$	9 ± 1
D_{s1}	$3 \le M^2 \le 6$	10 ± 1
B_1	$8 \le M^2 \le 12$	41 ± 1
B_{s1}	$9 \le M^2 \le 13$	43 ± 1

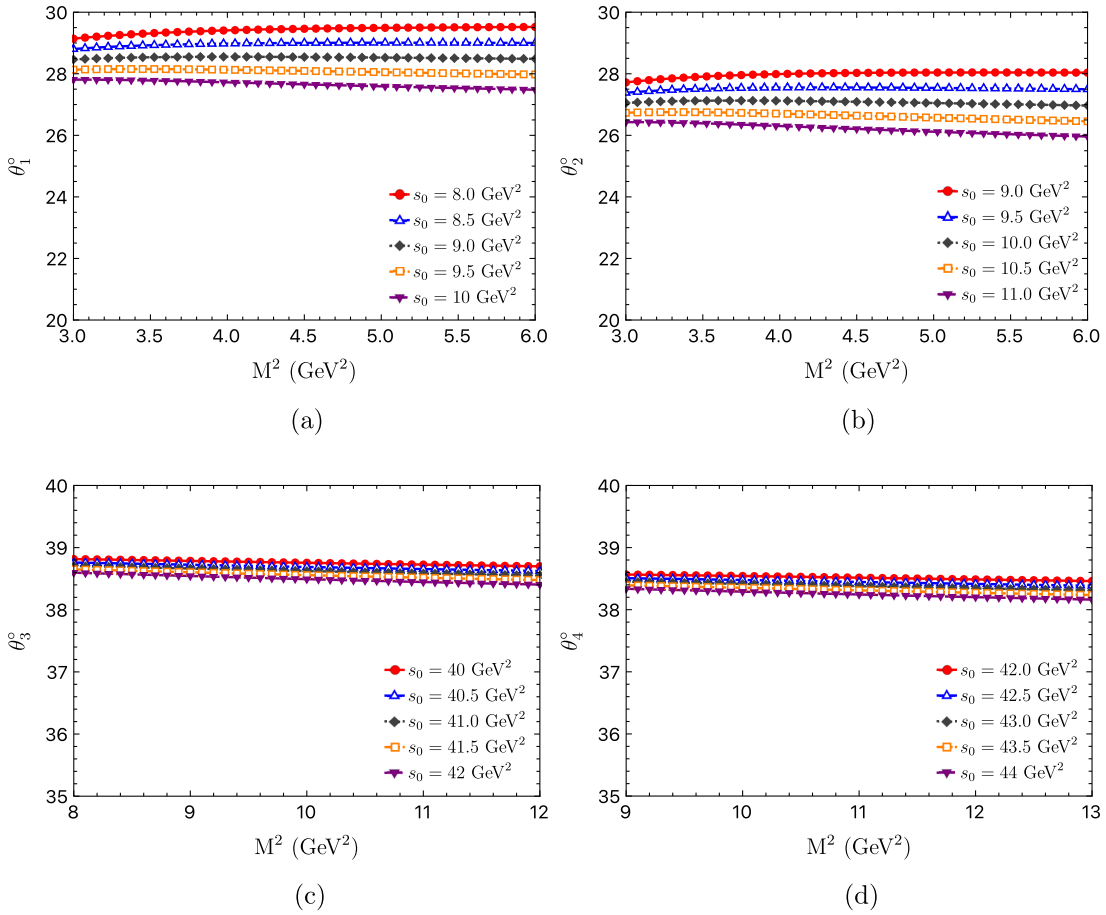


FIG. 1. Dependency of mixing angles on M^2 at several fixed values s_0: (a)$D_1 - D_1'$, (b)$D_{s1} - D_{s1}'$, (c) $B_1 - B_1'$, (d) $B_{s1} - B_{s1}'$.

40% of the total result. This condition can be expressed as

$$\frac{\int_{m_Q^2}^{s_0} ds\rho_1(s)e^{-s/M^2}}{\int_{m_Q^2}^{\infty} ds\rho_1(s)e^{-s/M^2}} \geq 0.6. \tag{14}$$

The lower limit of M^2 is determined by requiring that the operator product expansion should be convergent. Specifically, the contribution of the highest dimensional condensate must be less than 10% of the total result. The continuum threshold s_0 is selected to minimize the variation of the mixing angle within the Borel mass working region. These conditions lead to the following working regions for M^2 and s_0, as summarized in Table III.

Having determined the working regions of M^2 and s_0, the mixing angles θ between the axial-vector mesons $D_{1(s1)} - D'_{1(s1)}$ and $B_{1(s1)} - B'_{1(s1)}$ are evaluated. Their dependence on M^2 at several fixed s_0 values are presented in Fig. 1.

To account for uncertainties in the input parameters, as well as the auxiliary parameters M^2 and s_0, we performed a Monte Carlo analysis. By randomly sampling these parameters 5000 times within their respective uncertainty ranges, we generated statistical distributions for the mixing angles. The central values and uncertainties were extracted through Gaussian fits to these distributions (see Fig. 2).

The values of the mixing angles obtained from Monte Carlo analysis are collected in Table IV. For completeness, we present the predictions of other approaches existing in the literature.

Our results for $\theta_1 = 28.2 \pm 0.6°$, $\theta_2 = 26.6 \pm 0.6°$, $\theta_3 = 38.6 \pm 0.1°$, and $\theta_4 = 38.5 \pm 0.1°$ are notably close to the positive angle predicted by heavy quark effective theory $\theta = +35.3°$ [19]. The small deviations can be attributed to finite heavy quark mass corrections.

A key advantage of the method used in this work is that the mixing angles are determined only by QCD parameters and are free from hadronic degrees of freedom. This independence is particularly important, as many alternative approaches rely on meson mass inputs that are not yet well determined experimentally, introducing uncertainties into their predictions.

IV. CONCLUSION

In this study, we calculated the mixing angles between the axial-vector mesons $D_{1(s1)} - D'_{1(s1)}$ and $B_{1(s1)} - B'_{1(s1)}$

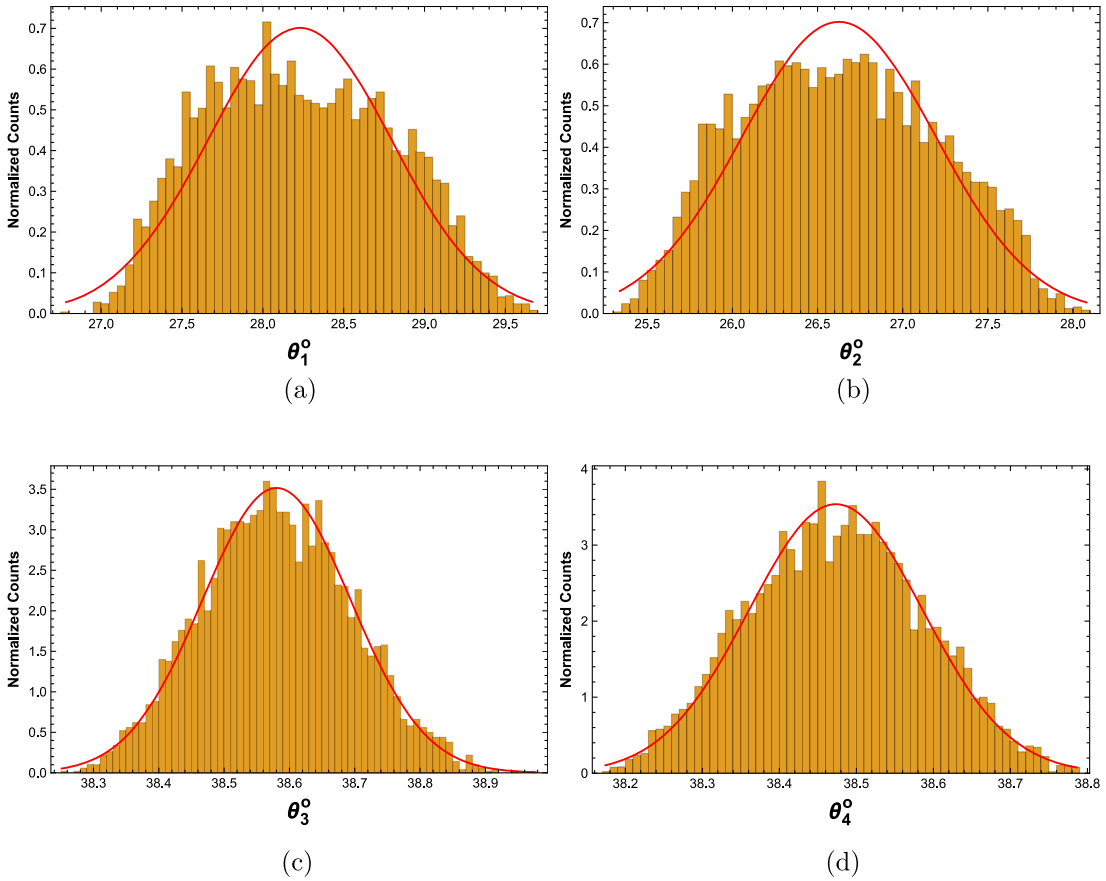


FIG. 2. Distribution of normalized counts for the mixing angles θ_1, θ_2, θ_3, and θ_4, obtained through a Monte Carlo analysis to determine their uncertainties. The input parameters were randomly varied within their uncertainties. The histograms represent the normalized counts, and the red curves show Gaussian fits to the resulting distributions: (a) $D_1 - D_1'$, (b) $D_{s1} - D_{s1}'$, (c) $B_1 - B_1'$, (d) $B_{s1} - B_{s1}'$.

within the framework of QCD sum rules. Our results are $\theta_1 = 28.2 \pm 0.6°$, $\theta_2 = 26.6 \pm 0.6°$, $\theta_3 = 38.6 \pm 0.1°$, and $\theta_4 = 38.5 \pm 0.1°$ and are in good agreement with the predictions of heavy quark effective theory in the heavy quark limit, for the mixing angle $\theta = 35.3°$. Furthermore, our results show compatibility with several existing studies

TABLE IV. Comparison of mixing angles between the heavy axial-vector mesons. Here θ_1, θ_2, θ_3, and θ_4 describe the mixing angles between the $D_{1(s1)} - D'_{1(s1)}$ and $B_{1(s1)} - B'_{1(s1)}$ states, respectively.

Reference	θ_1	θ_2	θ_3	θ_4
Present Work	$28.2 \pm 0.6°$	$26.6 \pm 0.6°$	$38.6 \pm 0.1°$	$38.5 \pm 0.1°$
[1]	…	…	$-35.2°$	$-39.6°$
[4]	$-25.7°$	$-37.5°$	…	…
[6]	$-58.3 \pm 9.0°$	…	…	…
[7]	$35.1°$	$-60.4°$	$-55.4°$	$-55.3°$
[9]	…	…	$-34.6°$	$-34.9°$
[10]	$29.0°$	$26.0°$	$31.7°$	$27.3°$
[11]	$43.5°$	$58.4°$	…	…
[12]	…	$-45.4°$	…	…
[14]	$34.0°$	$33.0°$	$35.0°$	$34.8°$
[15]	$25.7°$	$37.5°$	$30.3°$	$39.1°$
[16]	$-54.7°(35.3°)$	$-54.7°(35.3°)$	$-54.7°(35.3°)$	$-54.7°(35.3°)$
[3]	$-26.0°$	$-38.0°$	$-31.0°$	$-40.0°$
[5]	…	…	$30.3°(43.6°)$	$39.1°(37.9°)$
[8]	…	…	$-73.5 \pm 3.5°(-36.5 \pm 3.5°)$	…
[13]	$-55°$	$-55°$	…	…

in the literature. A key advantage of our approach is its independence from hadronic degrees of freedom.

Our predictions for the mixing angles can be tested through the analysis of semileptonic decays of B and B_c mesons. Specifically, decays such as $B_c \to B_1 \ell\nu$, $B_c \to B_{s1}^0 \ell\nu$, $B_s \to D_{s1} \ell\nu$, and $B_s \to D'_{s1} \ell\nu$ offer promising areas for experimental verification at facilities such as LHCb and Belle II.

The discrepancies observed between our results and those obtained using other theoretical approaches highlight the need for further theoretical refinements and experimental investigations.

APPENDIX: THE EXPRESSION OF THE SPECTRAL DENSITIES

Spectral densities corresponding to $\boldsymbol{g}_{\mu\nu}$ structure in the polarization operator.

$$
\begin{aligned}
\rho_1^{AA} - \rho_1^{BB} = e^{-m_Q^2/M^2} \Bigg\{ & \frac{1}{2^9 3^4 M^{14}} m_Q^5 [4(\langle g^2 G^2\rangle)^2 m_Q^2 + 24 m_0^2 m_q m_Q \langle g^3 G^3\rangle] \langle \bar{q} q\rangle \\
& - \frac{1}{2^9 3^4 M^{12}} m_Q^3 [(52(\langle g^2 G^2\rangle)^2 - 72\langle g^3 G^3\rangle m_0^2) m_Q^2 \\
& + 48 m_0^2 m_q m_Q (4\langle g^3 G^3\rangle + \langle g^2 G^2\rangle m_Q^2)] \langle \bar{q} q\rangle \\
& + \frac{1}{2^8 3^3 M^{10}} m_Q \{2 m_Q [m_Q (14(\langle g^2 G^2\rangle)^2 - 51\langle g^3 G^3\rangle m_0^2 - 12\langle g^2 G^2\rangle m_0^2 m_Q^2) \\
& + 6 m_q \langle g^3 G^3\rangle (m_0^2 + 2 m_Q^2) - 48\langle g^2 G^2\rangle m_0^2 m_q m_Q^2]\} \langle \bar{q} q\rangle \\
& - \frac{1}{2^6 3^3 M^8} \{5\langle g^2 G^2\rangle - 6 m_Q [7 m_0^2 m_Q - 2 m_Q^3 - 2 m_q (6 m_0^2 - m_Q^2)]\} m_Q \langle g^2 G^2\rangle \langle \bar{q} q\rangle \\
& + \frac{1}{2^8 3^3 m_Q M^8} \{6 m_Q \langle g^3 G^3\rangle [23 m_0^2 m_Q - 8 m_Q^3 + m_q (30 m_0^2 + 8 m_Q^2)]\} \langle \bar{q} q\rangle \\
& - \frac{1}{144 M^6} \{\langle g^2 G^2\rangle [4 m_0^2 m_Q - 6 m_Q^3 - m_q (4 m_0^2 - 7 m_Q^2)] - 12 m_0^2 m_q m_Q^4\} \langle \bar{q} q\rangle \\
& + \frac{1}{288 m_Q M^6} \langle g^3 G^3\rangle [2 m_0^2 + 12 m_q m_Q + 7 m_Q^2] \langle \bar{q} q\rangle \\
& - \frac{1}{48 M^4} [2 m_Q (\langle g^2 G^2\rangle - 6 m_0^2 m_Q^2) - 3 m_q (\langle g^2 G^2\rangle - 4 m_0^2 m_Q^2)] \langle \bar{q} q\rangle \\
& + \frac{1}{192 m_Q M^4 \pi^2} \langle g^3 G^3\rangle (m_q m_Q^2 + 2\pi^2 \langle \bar{q} q\rangle) \\
& - \frac{1}{96 M^2 \pi^2} m_Q [\langle g^2 G^2\rangle m_q + 24\pi^2 (m_0^2 + 2 m_q m_Q) \langle \bar{q} q\rangle] \\
& - \frac{1}{384 m_Q M^2 \pi^2} \langle g^3 G^3\rangle (m_q + 2 m_Q) \\
& + \frac{1}{96 m_Q \pi^2} [\langle g^2 G^2\rangle (m_q + m_Q) - 24\pi^2 \langle \bar{q} q\rangle (m_0^2 - 2 m_q m_Q + 4 m_Q^2)] \\
& - \frac{1}{384 m_Q^3 \pi^2} \langle g^3 G^3\rangle (3 m_q + 8 m_Q) \Bigg\} \\
& - \int_{m_Q^2}^{s_0} ds e^{-s/M^2} \left\{ \frac{1}{32 s \pi^2} \langle g^2 G^2\rangle + \frac{3}{8 m_Q s \pi^2} \{(s - m_Q^2)[s m_Q - m_Q^3 - m_q (s + m_Q^2)]\} \right\}. \qquad \text{(A1)}
\end{aligned}
$$

$$
\begin{aligned}
\rho_1^{AB} = e^{-m_Q^2/M^2}\Bigg\{ & -\frac{1}{2^8 3^4 M^{12}} m_Q^3 [14(\langle g^2G^2\rangle)^2 m_Q^2 + 96\langle g^3G^3\rangle m_0^2 m_q m_Q]\langle \bar{q}q\rangle \\
& - \frac{1}{2^7 3^3 M^{10}} m_Q [-(22(\langle g^2G^2\rangle)^2 - 36\langle g^3G^3\rangle m_0^2) m_Q^2 \\
& - 7 m_0^2 m_q m_Q (17\langle g^3G^3\rangle + 4\langle g^2G^2\rangle m_Q^2)]\langle \bar{q}q\rangle \\
& - \frac{1}{2^6 3^3 M^8}\langle g^2G^2\rangle m_Q \{25\langle g^2G^2\rangle - 6 m_Q [5 m_0^2 m_Q - m_q(14 m_0^2 - 4 m_Q^2)]\}\langle \bar{q}q\rangle \\
& + \frac{1}{2^6 3^2 m_Q M^8}\langle g^3G^3\rangle [5(\langle g^2G^2\rangle + m_0^2 m_Q^2) - m_q m_Q (23 m_0^2 - 10 m_Q^2)]\langle \bar{q}q\rangle \\
& + \frac{1}{432 m_Q M^6}[3\langle g^2G^2\rangle(\langle g^2G^2\rangle - 6 m_0^2 m_Q^2) + m_q m_Q \langle g^2G^2\rangle(20 m_0^2 - 39 m_Q^2) + 72 m_0^2 m_q m_Q^5]\langle \bar{q}q\rangle \\
& + \frac{1}{2^6 3^3 m_Q^2 M^6}\langle g^3G^3\rangle [108 m_0^2 m_Q - m_q(20 m_0^2 + 81 m_Q^2)]\langle \bar{q}q\rangle \\
& - \frac{1}{96 m_Q^2 M^4} m_q [3\langle g^3G^3\rangle - 4 m_Q^2 (3\langle g^2G^2\rangle - 14 m_0^2 m_Q^2)]\langle \bar{q}q\rangle \\
& + \frac{1}{12 m_Q M^2}\{2\langle g^2G^2\rangle - 3 m_Q [4 m_0^2 m_Q - m_q(m_0^2 + 4 m_Q^2)]\}\langle \bar{q}q\rangle \\
& - \frac{1}{96 m_Q M^2 \pi^2}\langle g^3G^3\rangle (2 m_q - m_Q) \\
& - \frac{1}{96 m_Q^2 \pi^2}\{35\langle g^2G^2\rangle m_Q^2 - 2 m_q [\langle g^2G^2\rangle m_Q + 8\pi^2(m_0^2 + 9 m_Q^2)\langle \bar{q}q\rangle]\} \\
& - \frac{1}{96 m_Q^3 \pi^2}[\langle g^3G^3\rangle (5 m_q - 2 m_Q)]\Bigg\} \\
& + \int_{m_Q^2}^{s_0} ds e^{-s/M^2}\Bigg\{\frac{1}{48 s^2 \pi^2}\langle g^2G^2\rangle (4s - m_Q^2) - \frac{1}{384 m_Q^2 s^2 \pi^2}\langle g^3G^3\rangle (2s + m_Q^2) \\
& - \frac{1}{8\pi^2 m_Q^2 s^2}(s - m_Q^2)^2 (s^2 - 6 s m_q m_Q - m_Q^4)\Bigg\}. \qquad \text{(A2)}
\end{aligned}
$$

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